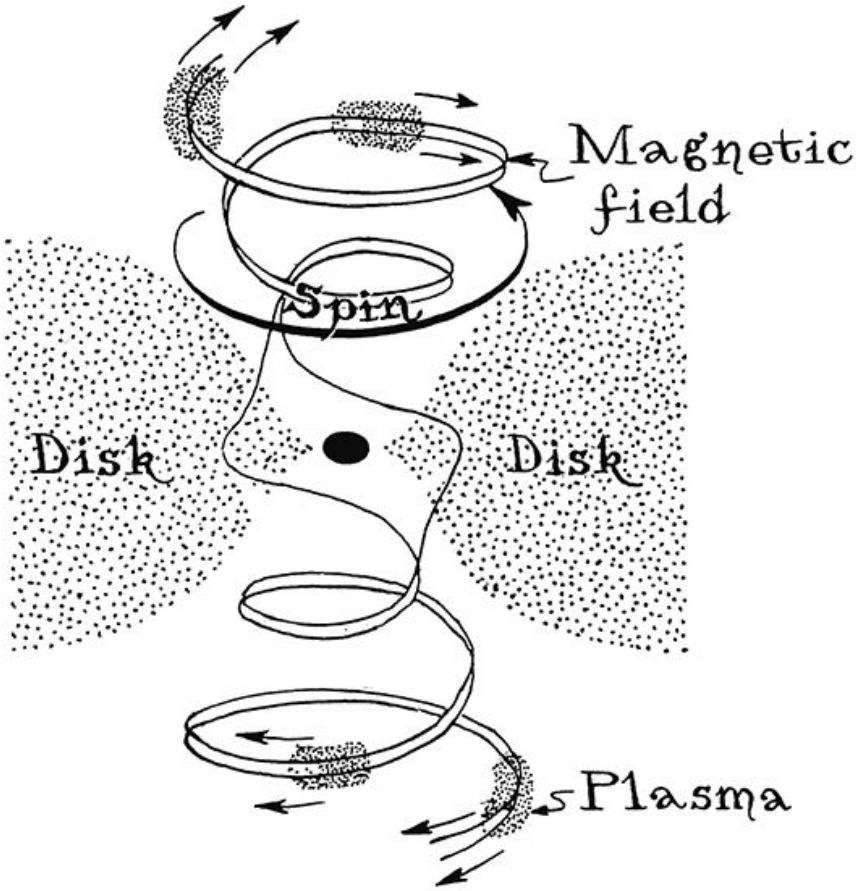


Fig. 9.4. Like Figure 9.3 but with magnetic field anchored in the accretion disk. *[Drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy.*]*

Whence Comes the Disk? Tidal Forces Tear Stars Apart

Lynden-Bell, in 1969, speculated that quasars live at the centers of galaxies. We don't see a quasar's host galaxy, he said, because its light is so much fainter than the quasar's light. The quasar drowns the galaxy out. In the decades since then, with improving technology, astronomers have found the galaxy's light around many quasars, confirming Lynden-Bell's speculation.

In those recent decades we also learned where most of the disk's gas comes from. Occasionally a star strays so close to the quasar's black hole that the hole's tidal gravity (Chapter 4) tears the star apart. Much of